

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

Let's break down the problem step by step to determine the total time it takes for Carla to download the file .

1. **Calculate the initial download time : **

- The file size is 200 GB .
- Carla 's normal download speed is 2 GB / minute .
- 40 % of the file size is :

$$\begin{aligned} &0.4 \times 200 \text{ GB} = 80 \text{ GB} \\ &\text{Time to download 80 GB at 2 GB / minute :} \\ &\frac{80 \text{ GB}}{2 \text{ GB / minute}} = 40 \text{ minutes} \end{aligned}$$

2. **Account for the forced restart and installation of updates : **

- The forced restart takes 20 minutes .

3. **Calculate the time to redownload the remaining portion : **

- The remaining 60 % of the file size is :

$$\begin{aligned} &0.6 \times 200 \text{ GB} = 120 \text{ GB} \\ &\text{Time to download 120 GB at 2 GB / minute :} \\ &\frac{120 \text{ GB}}{2 \text{ GB / minute}} = 60 \text{ minutes} \end{aligned}$$

4. **Sum up all the times : **

- Initial download time : 40 minutes
- Forced restart and updates : 20 minutes
- Redownload time : 60 minutes
- Total time :

$$40 \text{ minutes} + 20 \text{ minutes} + 60 \text{ minutes} = 120 \text{ minutes}$$

Thus , the total time it takes for Carla to download the file is 120 minutes .

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<think>
1. Initial download time : 40 minutes
2. Forced restart and updates : 20 minutes
3. Redownload time : 60 minutes
4. Total time : 40 + 20 + 60 = 120 minutes
</think>
<answer> 120 </answer>
```